

Generative AI Fundamentals

Knowledge-base document for the Domain-Specific RAG Chatbot

1. Introduction to Generative AI

Generative Artificial Intelligence (Generative AI) refers to AI systems that can create new content from patterns learned from data. The generated content may include text, images, audio, video, software code, or structured data.

Traditional machine-learning systems are often designed to predict a label, score, or numerical value. Generative systems instead focus on producing new outputs that follow patterns present in their training data.

Generative AI systems normally contain a trained model and an inference process. During inference, the model receives an input or prompt and produces an output according to the patterns encoded in its parameters.

2. Major Types of Generative AI

- Text generation: produces natural-language responses, summaries, explanations, and other written content.
- Image generation: produces images from textual or other conditioning inputs.
- Audio generation: produces speech, music, or other audio content.
- Video generation: produces or transforms sequences of visual content.
- Code generation: produces programming code, explanations, tests, or code transformations.

3. Generative AI Applications

Generative AI can support writing assistance, document summarization, conversational systems, content creation, software development, education, research assistance, and information retrieval.

In enterprise environments, generative systems can be combined with private document collections so that responses are based on organization-specific information rather than only general model knowledge.

4. Advantages

Generative AI can make interaction with information more natural, automate repetitive content creation, and provide explanations in conversational language.

When integrated with retrieval systems, it can help users locate and understand relevant information from large document collections.

5. Limitations and Challenges

Generative models may produce inaccurate or unsupported statements. This behavior is commonly called hallucination.

Model outputs can also depend on prompt quality, available context, training data, and model limitations. For high-stakes uses, generated information should be verified against reliable sources.

A retrieval-augmented system can reduce unsupported answers by supplying relevant source passages to the model and requiring the model to answer only from that supplied context.

6. Key Terms

Generative model: a model that produces new content.

Prompt: an input instruction or question provided to a generative model.

Inference: the process of using a trained model to produce an output.

Hallucination: an output that contains information not adequately supported by the available evidence or context.